Supplementary Material for "Fermi and Luttingers arcs: two concepts, realized on one surface"

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This supplementary material contains additional information to support the conclusions reached in the main text. Section I refers to the data availability of the publication. In Section II, we outline the computational details for the DΓA results. Section III provides additional DΓA results for a lower temperature ($\beta = 22.5/t$). In Section IV, the momentum-selective interaction of our proposed model Hamiltonian is discussed in relation to spin fluctuations of the Hubbard model. Section V details a qualitative picture of Luttinger zeros and the analytic structure of the Green's function, given different self-energies. This is connected to the self-energy of the DΓA solution for the Hubbard model at the nodal and antinodal points. In Section VI, we show that momentum cuts of the spectral function at the node (antinode) show a similar structure as qualitatively discussed in the previous section. Section VII discusses the different formation of Fermi arcs for positive or negative values of the hopping parameter $t'$, while section VIII treats the implications of the Fermi arcs on the loss of spectral weight and the reduction of the linear specific heat coefficient. In section IX, we discuss the specific high-temperature dependence of the model Hamiltonian, which arises due to the intrinsic momentum selectivity and in the last section X we compare its spectral function, Fermi, and Luttinger surface in the nodal Fermi arc regime for different values of interaction $\mathcal{V}$.

## I. DATA AVAILABILITY

A data set containing all numerical data and plot scripts used to generate the figures of this publication is publicly available at [S1].

## II. COMPUTATIONAL DETAILS

The dynamical mean-field calculations (DMFT), which are used as the starting point for the dynamical vertex approximation (DΓA), were performed using the continuous-time Quantum Monte Carlo in its hybridization expansion (CT-HYB) implemented in `w2dynamics` [S2]. For all quantities worm sampling was used and to ensure high-quality statistics we used about $10^9$ measurements. The DΓA calculations were performed using the `DGApy` [S3] framework. A $\boldsymbol{k}$-mesh of (140,140,1) grid points was used, to ensure high-quality momentum resolution. We used 60 and 80 positive Matsubara frequencies in the vertex function for $\beta = 12.5/t$ and $22.5/t$, respectively. The asymptotic treatment of the irreducible vertex is described in [S4] and we used 200, respectively 250 additional frequencies for the shell region. The $\lambda$ correction was performed both in the charge and the spin channel [S5, S6].

Analytic continuation was performed using the `ana_cont` package [S7]. We employed the "chi2kink" [S8] method for determining the hyperparameter $\alpha$. To avoid numeric instabilities of sharp delta-peaks we set a minimum imaginary part of the self-energy $\delta = -0.04$, i.e. we set $\Im(\Sigma_{\text{clip}}) = \min(\Im(\Sigma), \delta)$.

## III. DΓA AT LOWER TEMPERATURES

Below in Fig. S1 we show the same plot as in the main text, but for $\beta = 22.5/t$ instead of $\beta = 12.5/t$. As one would expect spectral features like the momentum-selective gap opening around the X point (antinode) are more pronounced.

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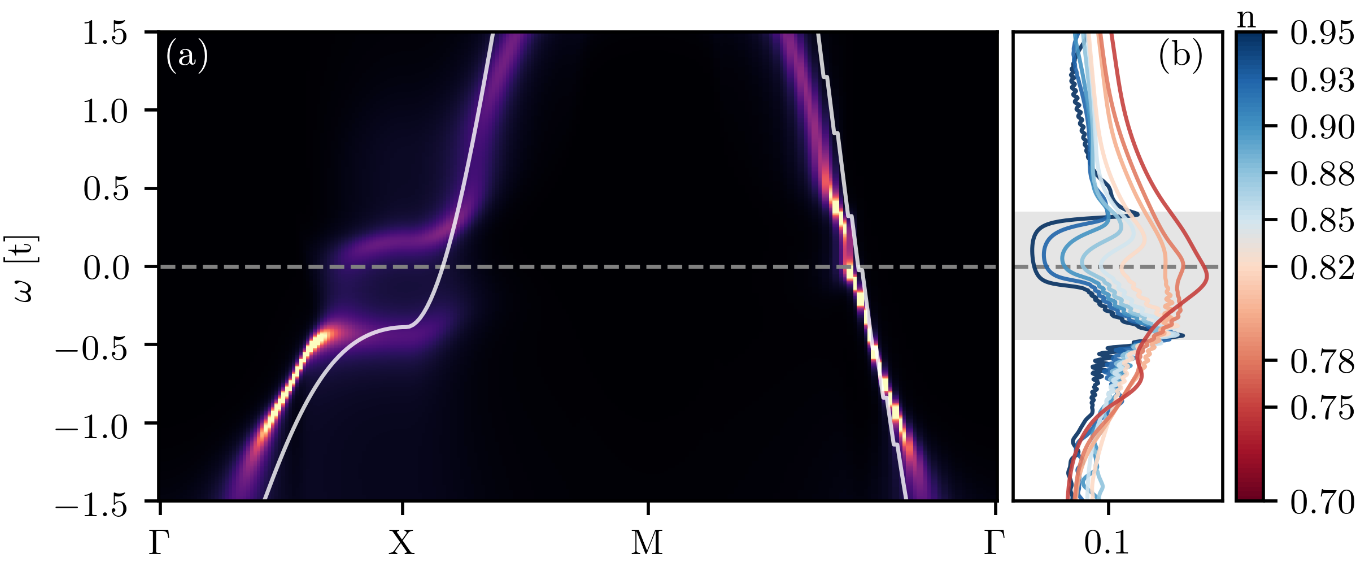


FIG. S1. Same as Fig. 2 in the main text, but for $\beta = 22.5/t$.

## IV. MOTIVATION FOR THE MOMENTUM-SELECTIVE INTERACTION

That the interaction term of our model in Eq. (1) of the main text is able to effectively mimic relevant features of the non-local correlations arising from the Hubbard interaction in the pseudogap regime is far from obvious. In fact, when taking a Fourier transform of the local Hubbard interaction to momentum space (cf. Fig. S2) one cannot directly see that certain interaction terms are more dominant than others a priori:

$$\frac{U}{2}\sum_{\sigma i} n_{\sigma i} n_{-\sigma i} = \frac{U}{2}\sum_{\boldsymbol{k}\boldsymbol{k}'\boldsymbol{q}} c^{\dagger}_{\sigma\boldsymbol{k}} c_{\sigma\boldsymbol{k}'} c^{\dagger}_{-\sigma\boldsymbol{k}'+\boldsymbol{q}} c_{-\sigma\boldsymbol{k}+\boldsymbol{q}} \tag{S1}$$

(note that we choose here the less used particle-hole transverse momentum notation, which is particular useful in the subsequent discussion). However, by considering the Feynman diagrammatic expansions for the square lattice—due to nesting and, possibly, also due to superexchange processes at intermediate-to-large $U$—we can expect that couplings of the form of $G^0_{\boldsymbol{k}} G^0_{\boldsymbol{k}+\boldsymbol{Q}} \frac{U}{2} G^0_{\boldsymbol{k}'} G^0_{\boldsymbol{k}'+\boldsymbol{Q}}$, $Q = (\pi, \pi)$, will yield the dominant contributions for low temperatures and near half-filling, especially in ladder diagrams of the spin channel. This can be readily seen, e.g., by performing a random phase approximation (RPA) for the Hubbard model. On the other hand, strong correlation effects are expected to play a key role in the description of the pseudogap and a perturbative method like RPA, where the irreducible vertex is kept fixed to the bare Hubbard interaction, is largely insufficient. The model we propose allows us to take a different approach, by restricting the interaction in momentum-space to $\boldsymbol{q} = \boldsymbol{Q}$, thereby considering the specific coupling corresponding to the dominant spin-fluctuations. Further, the interaction is restricted to $\delta_{\boldsymbol{k}\boldsymbol{k}'}$, inspired by the HK model, to make the model exactly solvable while still possibly incorporating strong correlation effects. Previous studies of the HK model demonstrated that interactions diagonal in momentum space, often allow for an analytical study of otherwise hard to tackle many-body physics phenomena, such as Mott like transitions, e.g. in Refs. [S9, S10]. In particular, Refs. [S11, S12] have explored the specific effects of a $\boldsymbol{k}$-dependent modification of the HKM interaction.

Fig. S3 provides an illustrative comparison of the energy levels coupled by the interaction in the proposed model (left) and the Hubbard model (right).

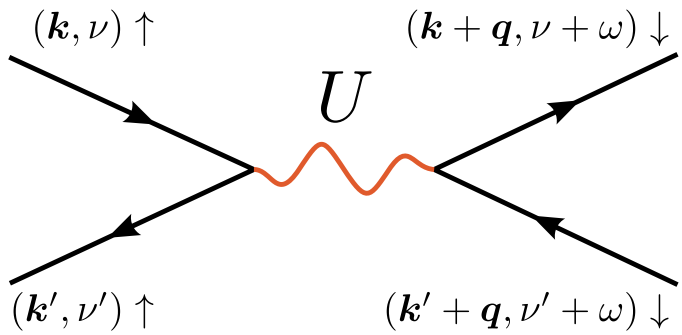


FIG. S2. Feynman-diagrammatic representation of the Hubbard interaction in $k$-space. For the proposed model, only $\boldsymbol{q} = \boldsymbol{Q} = (\pi, \pi)$ and momenta $\boldsymbol{k} = \boldsymbol{k}'$ contributions are considered, the full frequency dependence of the interaction $(\nu, \nu', \omega)$ remains.

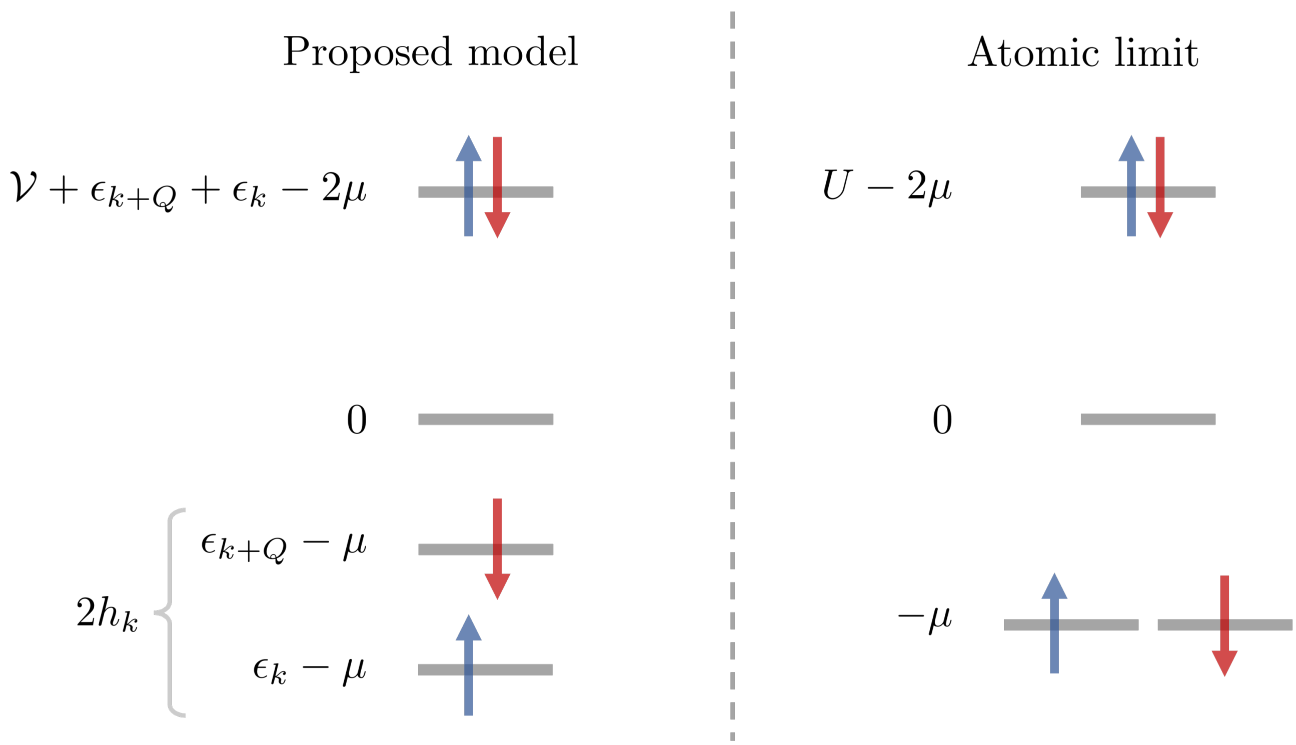


FIG. S3. Schematic comparison between the energy levels of the proposed model and the atomic limit. The energy levels of the proposed model (left), illustrate that the coupling between $\boldsymbol{k}, \uparrow$ and $\boldsymbol{k}+\boldsymbol{Q}, \downarrow$ costs an additional energy $\mathcal{V}$ if both momenta are occupied. The Hubbard interaction $U$ (right) affects a double-occupied single site. This sketch refers to the specific situation of $\mathcal{V} > \epsilon_{k+Q} + \epsilon_k - 2\mu > 0 > \epsilon_{k+Q} + \mu > \epsilon_k + \mu$ for the proposed model as well as for $U/2 > \mu > 0$ in the atomic limit of the Hubbard model.

## V. LUTTINGER ZEROS - A QUALITATIVE PERSPECTIVE

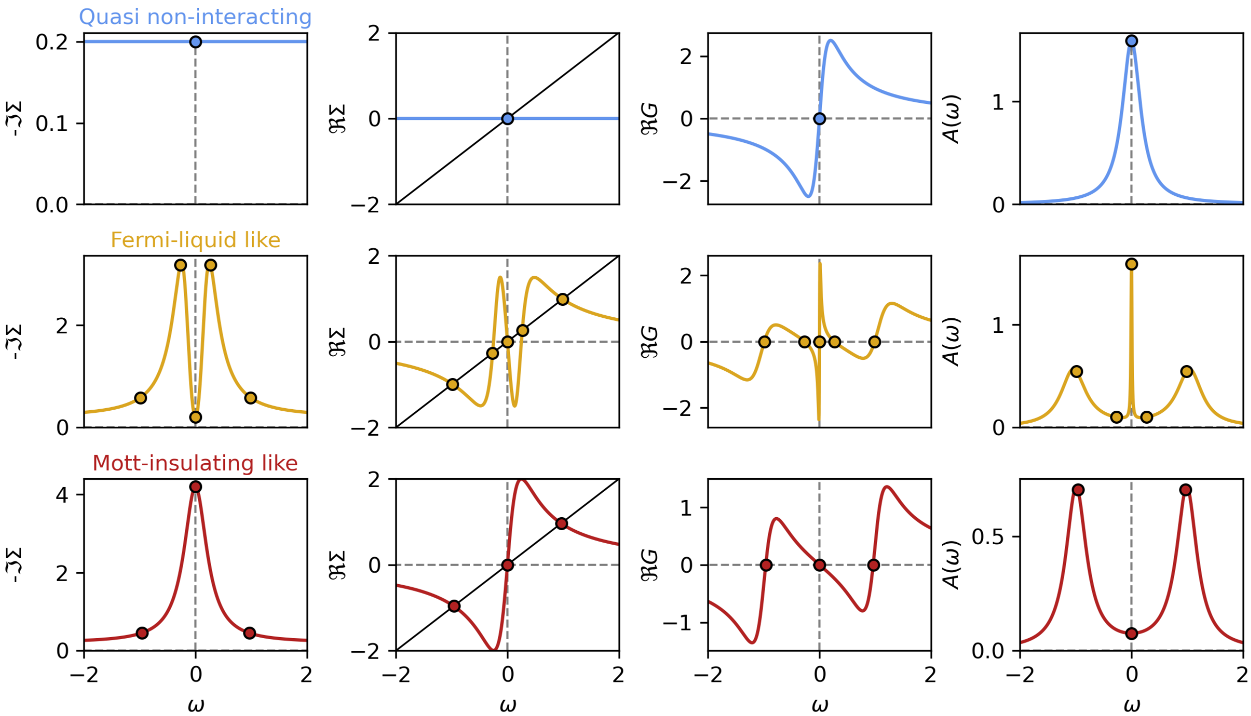


FIG. S4. Influence of the functional form of the self-energy on the structure of the Green's function at a Fermi vector $\mathbf{k}_\mathrm{F}$. From left to right: minus the imaginary part of the self-energy $-\Im\Sigma$, the real part of the self-energy $\Re\Sigma$, the real part of the Green's function $\Re G$, the spectral function $A(\omega)$. From top to bottom: self-energy from Eq. (S4) mimicking an effectively non-interacting system, self-energy from Eq. (S5) mimicking a renormalized metal, self-energy from Eq. (S6) mimicking an interaction driven insulator. The black line in the second column corresponds to $\omega - \epsilon(\mathbf{k}_\mathrm{F}) + \mu$ and the intersection with the real part of the self-energy determines the energy location of the "bands". Whether or not these "bands" are also visible in the spectral function depends on the magnitude of $\Im\Sigma$.

To better understand how zeros of the Green's function arise in the presence of interactions, we discuss three scenarios in Fig. S4. Let us note that the discussion below is qualitative in nature and only serves to illustrate how different functional forms of the self-energy $\Sigma$ change the spectral function $A(\omega)$, which is accessible in ARPES experiments. Furthermore, we discuss the appearance of zeros in the Green's function.

The single-particle Green's function is given by

$$G(\omega, \mathbf{k}) = \frac{1}{\omega - \epsilon(\mathbf{k}) + \mu - \Sigma(\omega, \mathbf{k})}, \tag{S2}$$

where $\epsilon(\mathbf{k})$ is the energy-momentum dispersion and $\mu$ the chemical potential. For the following, we consider the Green's function at a single Fermi vector $\mathbf{k}_\mathrm{F}$, defined as a momentum vector that satisfies

$$\omega - \epsilon(\mathbf{k}_\mathrm{F}) + \mu - \Re\Sigma(\omega, \mathbf{k}_\mathrm{F}) = 0, \tag{S3}$$

at $\omega = 0$.

To best understand the influence of the self-energy on the Green's function, we plot the corresponding (minus) imaginary part of the self-energy ($-\Im\Sigma$) in the first row, the real part ($\Re\Sigma$) in the second, the real part of the Green's function ($\Re G$) in the third and finally the spectral function ($A(\omega) = -\frac{1}{\pi}\Im G$) in the rightmost column.

In the first scenario (first row in blue), the scattering rate is constant ($\Delta_0$) with a high-frequency cut-off

$$\Im\Sigma = -\Delta_0 \;\; \Theta(\omega^* - |\omega|), \tag{S4}$$

where $\Delta_0$ is a positive real constant and $\omega^*$ is some frequency larger than the bandwidth. This essentially describes the *quasi* non-interacting case with some finite lifetime of the states, e.g. due to impurity scattering [S13]. The resulting spectrum is a single Lorentzian peak centered around the Fermi energy (here $\omega = 0$), where the real part of the Green's function changes sign (marked by a dot).

For the second case, the self-energy is inspired by that of Fermi-liquids, or renormalized metals,

$$\Im\Sigma = -\Delta_1 \ \frac{1+(\sigma\omega)^2}{1+(\sigma\omega)^4} - \Delta_0 \ \Theta(\omega^* - |\omega|), \tag{S5}$$

with $\sigma$, $\Delta_0$, and $\Delta_1$ positive real constants. In addition to the constant scattering rate of Eq. (S4) a term which scales as $\omega^2$ for small frequencies, reminiscent of Fermi-liquids, has been added. The $1/\omega^4$ ensures the correct high-frequency asymptotics of the self-energy and the real part has been computed using the Kramers-Kronig relations [S14, S15].

The resulting spectrum, center row (orange) in Fig. S4, displays a three-peak structure, similar to DMFT solutions for the correlated metallic regime of the square-lattice Hubbard model, [S5, S16]. Already this simple self-energy is enough for the appearance of zeros in the Green's function, which separate the three "poles". Both poles and zeros correspond to solutions of Eq. (S3), which is indicated by the intersection of the black line ($\omega - \epsilon(\mathbf{k}_\mathrm{F}) + \mu$) and $\Re\Sigma$ (colored line). The difference (w.r.t. finite background scattering) is that the poles correspond to a small $\Im\Sigma$ and large spectral weight, similar to the quasiparticle and Hubbard side-bands in DMFT. At the zeros of the Green's function, however, spectral weight is notably absent and they are consequently invisible in ARPES.

For the "Fermi-liquid-like" case these zeros are located away from the Fermi energy. However, in interaction-driven insulators, e.g. Mott insulators, these zeros can be directly at the Fermi energy [S17]. Eq. (S6) displays a simple self-energy, which replicates this behavior,

$$\Im\Sigma = -\Delta_1 \ \frac{1}{1+(\sigma\omega)^2} - \Delta_0 \ \Theta(\omega^* - |\omega|), \tag{S6}$$

where the crucial part is that the self-energy has a peak at (or close to) the Fermi energy. This suppresses the spectral weight at the Fermi energy, which becomes a Luttinger zero. At the same time, the corresponding (frequency dependent) real part of the self-energy creates two additional solutions for Eq. (S3), which correspond to the two peaks visible in the spectrum above and below the Fermi energy.

This qualitative picture outlines what is happening to the Fermi surface in the pseudogap: In the hole-doped case, the self-energy at the node is qualitatively similar to the "Fermi-liquid-like" one discussed here and the corresponding spectrum appears metallic, yielding the Fermi arcs. At the antinode, however, the self-energy developes a peak close to the Fermi surface, resulting in a gapped spectrum with a Luttinger zero: the Luttinger arc, which is invisible in ARPES. For the electron-doped case, this picture is simply reversed.

In the following, we compare this qualitative picture to the results of the model Hamiltonian in Eq. (1) of the main text and to the DΓA solution of the Hubbard model, both for the nodal Fermi arc at hole doping. For the model in Eq. (1) the situation for $\mathcal{V} = t$, $n = 0.85$, $\beta = 12.5/t$ is shown in Fig. S5. For the plot on the real frequency axis, where $\mathrm{i}\nu_n \to \omega + \mathrm{i}\delta$, we use a spectral broadening of $\delta = 0.1$. Here, at the node (top row), the vanishingly small self-energy leaves the corresponding spectral function almost unaltered w.r.t. the non-interacting solution, reflecting the fact that the model preserves $\hat{n}_\mathbf{k}$ as a good quantum number. As a matter of course, there are no Hubbard bands. In contrast, for the antinode (bottom row), the self-energy clearly displays the spectral behavior of an interaction-driven insulator.

Fig. S6 displays this behavior for the DΓA solution of the Hubbard model at $n = 0.85$ and $\beta = 12.5/t$. For the node (left in blue) the self-energy (b) shows a trough at the Fermi energy ($\omega = 0$), which results in a metallic spectrum (a). On the contrary, the self-energy at the antinode develops a peak close to the Fermi energy (d), which results in a two-peak structure in the spectrum (b).

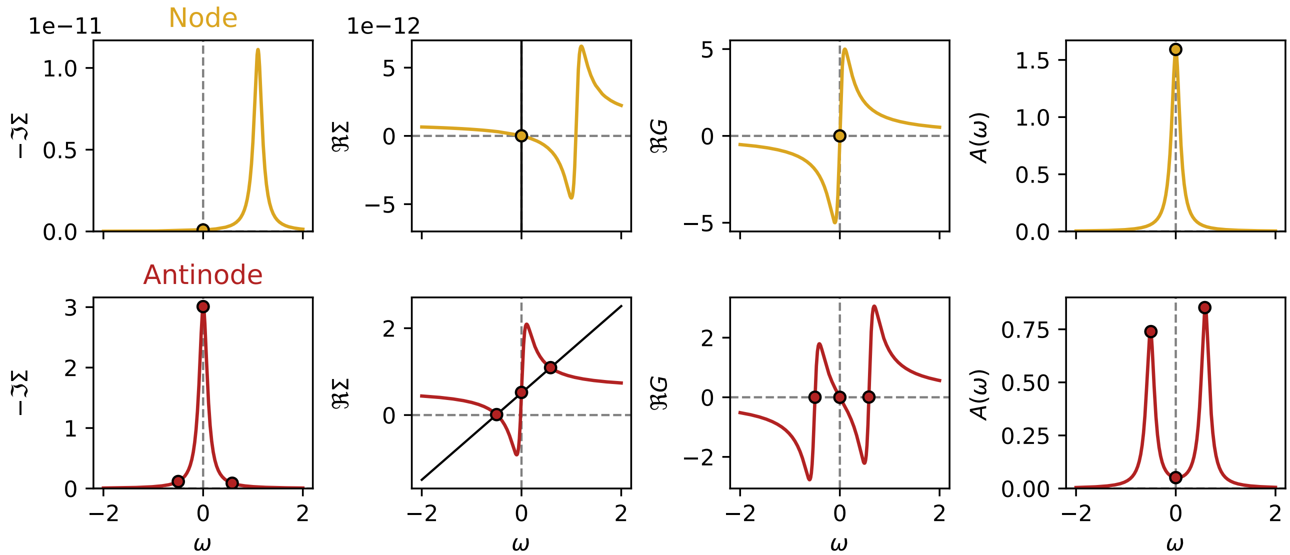


FIG. S5. Similar to Fig. S4, showing the difference of the functional form between solutions on the node (Fermi arc) and antinode (Luttinger arc) for the proposed model in the nodal Fermi arc regime at $\mathcal{V} = 1.1t$, $n = 0.85$, $\beta = 12.5/t$, on the real frequency axis, where $\mathrm{i}\nu_n \to \omega + \mathrm{i}\delta$ with spectral broadening $\delta = 0.1$. From left to right: minus the imaginary part of the self-energy $-\Im\Sigma$, the real part of the self-energy $\Re\Sigma$, the real part of the Green's function $\Re G$, and the spectral function $A(\omega)$. Top: self-energy at the node showing an almost non-interacting solution. Bottom: self-energy at the antinode resembling an interaction-driven insulator.

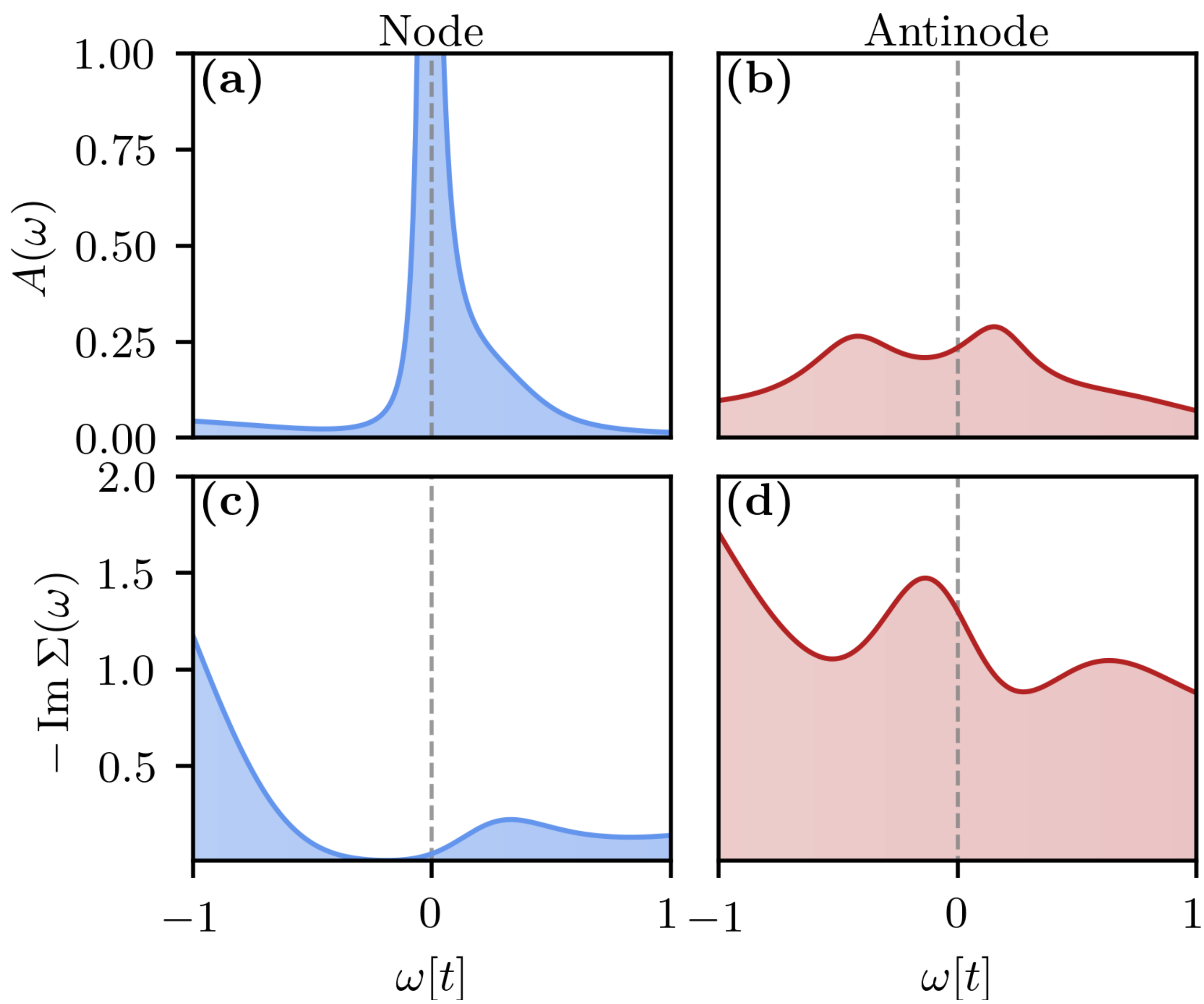


FIG. S6. Spectrum (top) and self-energy (bottom) at the node (left) and antinode (right) for the DΓA solution of the Hubbard model at $n = 0.85$, $\beta = 12.5/t$ and $U = 8t$.

## VI. SPECTRUM ALONG A HIGH-SYMMETRY PATH IN THE BRILLOUIN ZONE

As supplemental information to Fig. 3 in the main text, we plot the spectral function for the same parameters along a high-symmetry path (Γ-X-M-Γ) below. The left side shows the DΓA solution of the Hubbard model and the right side the exact solution of the proposed model. The lower panels show the spectral function and the real part of the single-particle Green's function along cuts at the node and antinode respectively.

The momentum-selective insulating behavior in the pseudogap is clearly visible as a double-peak (gapped) structure at the antinode, while the node displays a single peak corresponding to the Fermi arc. Furthermore, the real part of the Green's function contains information about the location of the Luttinger surface, as a zero-crossing without spectral weight and is observed within the gap.

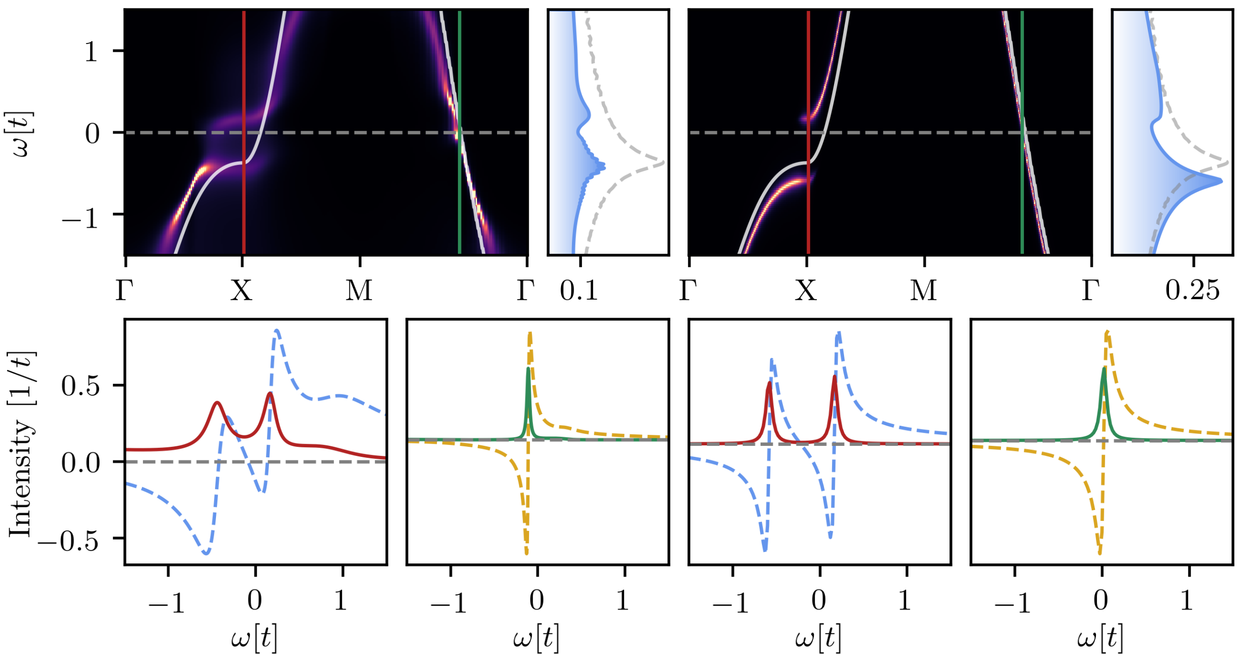


FIG. S7. Top row: $A(\omega, k)$ along a high-symmetry path, $\Gamma = (0,0)$, X$= (\pi, 0)$ and M$= (\pi,\pi)$, in the Brillouin zone for $n = 0.85$, $\beta = 22.5/t$, and corresponding momentum-integrated spectral function. The white lines display the non-interacting band dispersion for the same filling and the dashed gray lines show the corresponding non-interacting momentum-integrated spectral function. Left: DΓA solution of the Hubbard model at $U = 8t$. Right: analytic solution of the proposed model at $\mathcal{V} = 1.1t$. Lower panels below the top panels display respective momentum cuts of the spectral function (solid) and real part of the Green's function (dashed) for two momenta at the X point (left) and the node (right).

## VII. FERMI AND LUTTINGER ARCS FOR POSITIVE $t'$

In the main text we discussed the spectral function and Fermi/Luttinger arc properties for the tight-binding (tb) parameter set $t = 1$, $t' = -0.2$ and $t'' = 0.1$, inspired by Wannier models for cuprates [S5] and previous studies [S18, S19]. However, our general description of the mechanism and formation of Fermi and Luttinger arcs is valid for any set of tb parameters we tested. Below in Fig. S8, we illustrate this for $t' = +0.2$ and $t'' = -0.1$, i.e. inverting the sign of the next and next-next nearest neighbor hopping. Our results show again Fermi and Luttinger arcs, with the observation of Fermi arcs at the antinode (node), for electron- (hole-) doping. The remainder of the tight-binding Fermi surface becomes gapped and transforms into a Luttinger arc.

This also illustrates, that for our proposed model and differently to DCA calculations for the Hubbard model [S20], the existence of a pseudogap does not appear to be affected by the electron/hole nature of the underlying tb FS. That is, in our proposed model, a pseudogap can form for both electron- and hole-like tb parameters, provided that the interaction strength is large enough and the tb FS crosses the AFZB. The respective location of the Fermi and Luttinger arcs in momentum space depends on both the sign of $t'$ and whether the system is hole or electron doped. We summarize this briefly in Table I below.

On the other end (at least for $t''=0$), we note that the occurrence of the pseudogap is directly associated with a change in the nature of the tb FS. This is, because such change is connected to the tb FS crossing of the AFZB. As a result, for $t' > 0$ a pseudogap appears only for electron-like FSs, while for $t' < 0$ only for hole-like ones.

TABLE I. Location of the Luttinger arc (pseudogap) in spectrum of the proposed model for different signs of $t'$ and dopings.

| - | negative $t'$ ($t' < 0$) | positive $t'$ ($t' > 0$) |
|---|---|---|
| hole doping | antinode | node |
| electron doping | node | antinode |

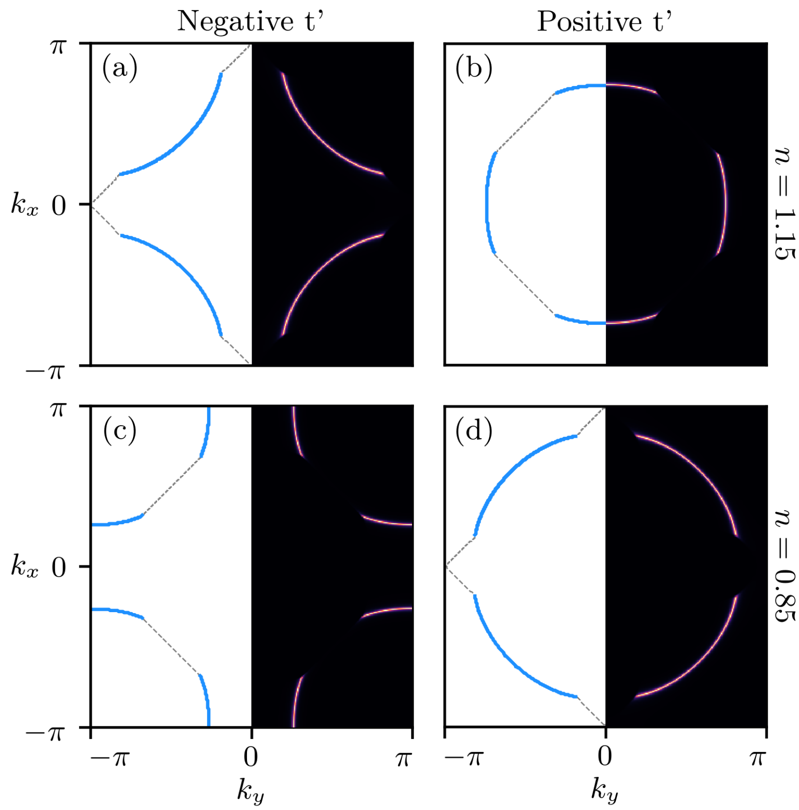


FIG. S8. Fermi surface (blue line) and Luttinger surfaces (gray line) computed for the proposed model at hole- (top row) and electron-doping (bottom row) at $\beta = 50/t$ and $\mathcal{V} = 1.1t$. Left column: $t = 1$, $t' = -0.2$ and $t'' = +0.1$. Right column: $t = 1$, $t' = +0.2$ and $t'' = -0.1$ The left half of the Brillouin zone displays the Fermi surface (blue line) and Luttinger surface (gray dashed), while the right half shows the spectral function at zero frequency ($A(\omega = 0)$).

## VIII. LOSS OF SPECTRAL WEIGHT AND IMPLICATIONS FOR THE SPECIFIC HEAT

The model Hamiltonian in Eq. (1) of the main text allows for a transparent view on the evolution of the disconnected Fermi surfaces (Fermi arcs) as function of doping. In Fig. S9 the spectral weight at the Fermi level

$$A(\omega = 0) = 2\sum_{\boldsymbol{k}} \left[(1 - n_{\boldsymbol{k}+\boldsymbol{Q}})\delta(\xi_{\boldsymbol{k}}) + n_{\boldsymbol{k}+\boldsymbol{Q}}\delta(\xi_{\boldsymbol{k}} + \mathcal{V})\right] \tag{S7}$$

as function of the electron density $n$ for different interaction strength $\mathcal{V}$ is displayed for $T \to 0$. For the regime, where the Fermi surface (FS) becomes disconnected and evolves into a Luttinger surface (LS) (gray-shaded area) the reduced FS leads to a corresponding loss of spectral weight at the Fermi level in comparison to the non-interacting $\mathcal{V} = 0$ case.

In our model, the major loss of spectral weight at the Fermi level is directly reflecting the reduction of the FS (i.e., the length of the Fermi arc). In the low-$T$ behavior of the electronic specific heat $c_V = \frac{\partial \langle H \rangle}{\partial T}$ a similar effect can be observed, where

$$\begin{aligned} \langle H \rangle &= \sum_{\boldsymbol{k}} \frac{1}{Z_{\boldsymbol{k}}} \left[2\xi_{\boldsymbol{k}}\, \mathrm{e}^{-\beta\xi_{\boldsymbol{k}}} + (\xi_{\boldsymbol{k}} + \xi_{\boldsymbol{k}+\boldsymbol{Q}} + \mathcal{V})\, \mathrm{e}^{-\beta(\xi_{\boldsymbol{k}}+\xi_{\boldsymbol{k}+\boldsymbol{Q}}+\mathcal{V})}\right] \\ &= \sum_{\boldsymbol{k}} \left[2\xi_{\boldsymbol{k}} f_{\xi_{\boldsymbol{k}}}(1 - n_{\boldsymbol{k}+\boldsymbol{Q}}) + 2(\xi_{\boldsymbol{k}} + \mathcal{V}) f_{\xi_{\boldsymbol{k}}+\mathcal{V}} n_{\boldsymbol{k}+\boldsymbol{Q}} - \mathcal{V} d_{\boldsymbol{k}}\right], \end{aligned} \tag{S8}$$

$Z_{\boldsymbol{k}} = 1 + \mathrm{e}^{-\beta\xi_{\boldsymbol{k}}} + \mathrm{e}^{-\beta\xi_{\boldsymbol{k}+\boldsymbol{Q}}} + \mathrm{e}^{-\beta(\xi_{\boldsymbol{k}}+\xi_{\boldsymbol{k}+\boldsymbol{Q}}+\mathcal{V})}$, $n_{\boldsymbol{k}+\boldsymbol{Q}} = \frac{\mathrm{e}^{-\beta\xi_{\boldsymbol{k}+\boldsymbol{Q}}} + \mathrm{e}^{-\beta(\xi_{\boldsymbol{k}}+\xi_{\boldsymbol{k}+\boldsymbol{Q}}+\mathcal{V})}}{Z_k}$, $d_{\boldsymbol{k}} = \frac{\mathrm{e}^{-\beta(\xi_{\boldsymbol{k}}+\xi_{\boldsymbol{k}+\boldsymbol{Q}}+\mathcal{V})}}{Z_k}$ and $f_\xi = \frac{1}{\mathrm{e}^{\beta\xi}+1}$ is the Fermi function.

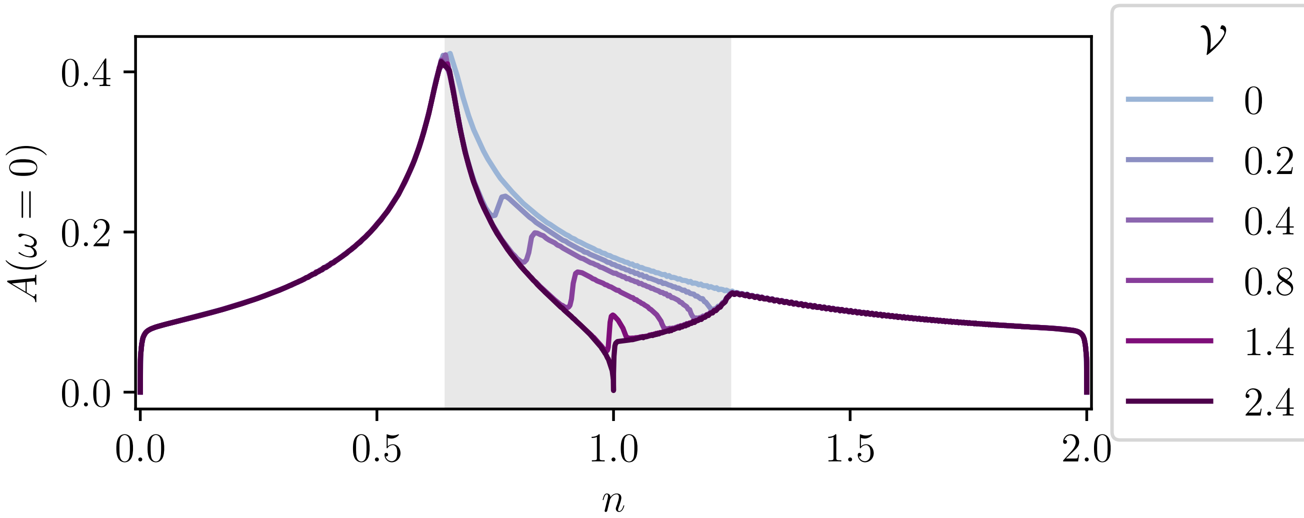


FIG. S9. Spectral weight at the Fermi level $A(\omega = 0)$ for different interaction strength and electron density for $T \to 0$ ($1/T = 10^6/t$). In comparison with $\mathcal{V} = 0$, the disconnected Fermi surface (gray-shaded area) leads to a depletion of spectral weight.

Taking the derivative of Eq. (S8) with respect to the temperature $T$, we get

$$c_V = \overbrace{-2\beta \sum_{\boldsymbol{k}} \left[\xi_{\boldsymbol{k}}^2 f'_{\xi_{\boldsymbol{k}}}(1 - n_{\boldsymbol{k}+\boldsymbol{Q}}) + (\xi_{\boldsymbol{k}} + \mathcal{V})^2 f'_{\xi_{\boldsymbol{k}}+\mathcal{V}} n_{\boldsymbol{k}+\boldsymbol{Q}}\right]}^{c_V^{FS}} \underbrace{-\beta^2 \sum_{\boldsymbol{k}} \left[2\frac{\partial n_{\boldsymbol{k}+\boldsymbol{Q}}}{\partial \beta}(-\xi_{\boldsymbol{k}} f_{\xi_{\boldsymbol{k}}} + (\xi_{\boldsymbol{k}} + \mathcal{V}) f_{\xi_{\boldsymbol{k}}+\mathcal{V}}) - \mathcal{V}\frac{\partial d_{\boldsymbol{k}}}{\partial \beta}\right]}_{c_V^{LS}}, \tag{S9}$$

where the first term on the right hand side of Eq. (S9) entails derivatives of the Fermi function and gives the usual contribution of the FS ($c_V^{FS} \propto \gamma T$ for $T \to 0$ ) while the second term yields a “thermally activated” behavior associated to the LS ($c_V^{LS}$), with

$$\begin{aligned} c_V^{LS} = -\beta^2 \sum_{\boldsymbol{k}} \big[ & \mathcal{V}^2 d_{\boldsymbol{k}}(d_{\boldsymbol{k}} + 1 - f_{\xi_{\boldsymbol{k}}+\mathcal{V}} - f_{\xi_{\boldsymbol{k}+\boldsymbol{Q}}+\mathcal{V}}) \\ & + 2\mathcal{V} d_{\boldsymbol{k}} \left((1 - n_{\boldsymbol{k}+\boldsymbol{Q}})\xi_{\boldsymbol{k}}(f_{\xi_{\boldsymbol{k}}} - f_{\xi_{\boldsymbol{k}}+\mathcal{V}}) + (1 - n_{\boldsymbol{k}})\xi_{\boldsymbol{k}+\boldsymbol{Q}}(f_{\xi_{\boldsymbol{k}+\boldsymbol{Q}}} - f_{\xi_{\boldsymbol{k}+\boldsymbol{Q}}+\mathcal{V}})\right) \\ & + n_{\boldsymbol{k}+\boldsymbol{Q}}(1 - n_{\boldsymbol{k}+\boldsymbol{Q}})(f_{\xi_{\boldsymbol{k}}} - f_{\xi_{\boldsymbol{k}}+\mathcal{V}})(\xi_{\boldsymbol{k}}\xi_{\boldsymbol{k}+\boldsymbol{Q}} - \xi_{\boldsymbol{k}}^2(f_{\xi_{\boldsymbol{k}}} - f_{\xi_{\boldsymbol{k}}+\mathcal{V}})) \\ & + n_{\boldsymbol{k}}(1 - n_{\boldsymbol{k}})(f_{\xi_{\boldsymbol{k}+\boldsymbol{Q}}} - f_{\xi_{\boldsymbol{k}+\boldsymbol{Q}}+\mathcal{V}})(\xi_{\boldsymbol{k}}\xi_{\boldsymbol{k}+\boldsymbol{Q}} - \xi_{\boldsymbol{k}+\boldsymbol{Q}}^2(f_{\xi_{\boldsymbol{k}+\boldsymbol{Q}}} - f_{\xi_{\boldsymbol{k}+\boldsymbol{Q}}+\mathcal{V}}))\big] \end{aligned} \tag{S10}$$

(cf. Fig. S10). Hence, in the regime of the disconnected FS, the reduced length of the Fermi arc leads to an effectively *reduced* coefficient $\gamma$ in the specific heat $c_V(T)$ (see Fig. S11).

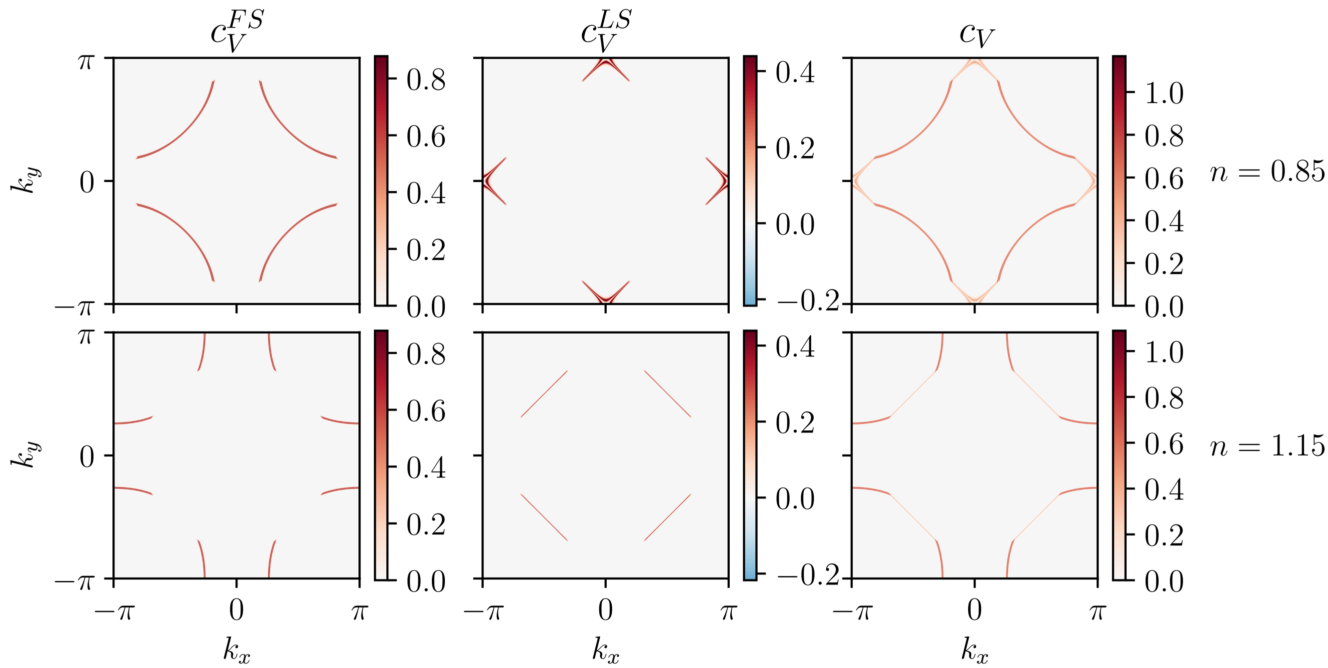


FIG. S10. $k$-resolved contributions to the electronic specific heat $c_V = c_V^{FS} + c_V^{LS}$ (right column) divided into the Fermi surface $c_V^{FS}$ (left column) and Luttinger surface $c_V^{LS}$ (middle column) contributions for hole (top row) and electron (bottom row) doping at interaction strength $\mathcal{V} = 2.4t$ and temperature $T = 1/100\,t$.

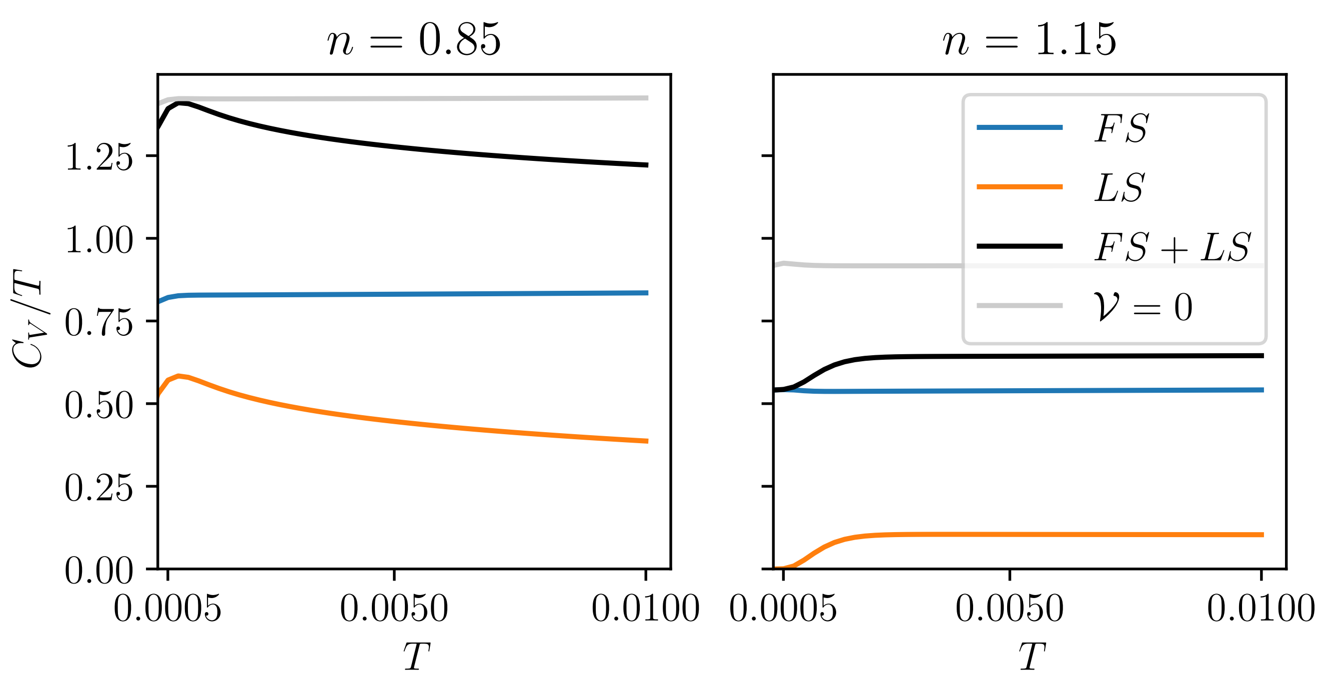


FIG. S11. Temperature dependence of the different contributions to the electronic specific heat $c_V/T = c_V^{FS}/T + c_V^{LS}/T$ separated into the Fermi surface $FS$ and Luttinger surface $LS$ contributions for hole (left panel) and electron (right panel) doping with interaction strength $\mathcal{V} = 2.4t$. The specific heat of the corresponding non-interacting model ($\mathcal{V} = 0$) is shown as a comparison.

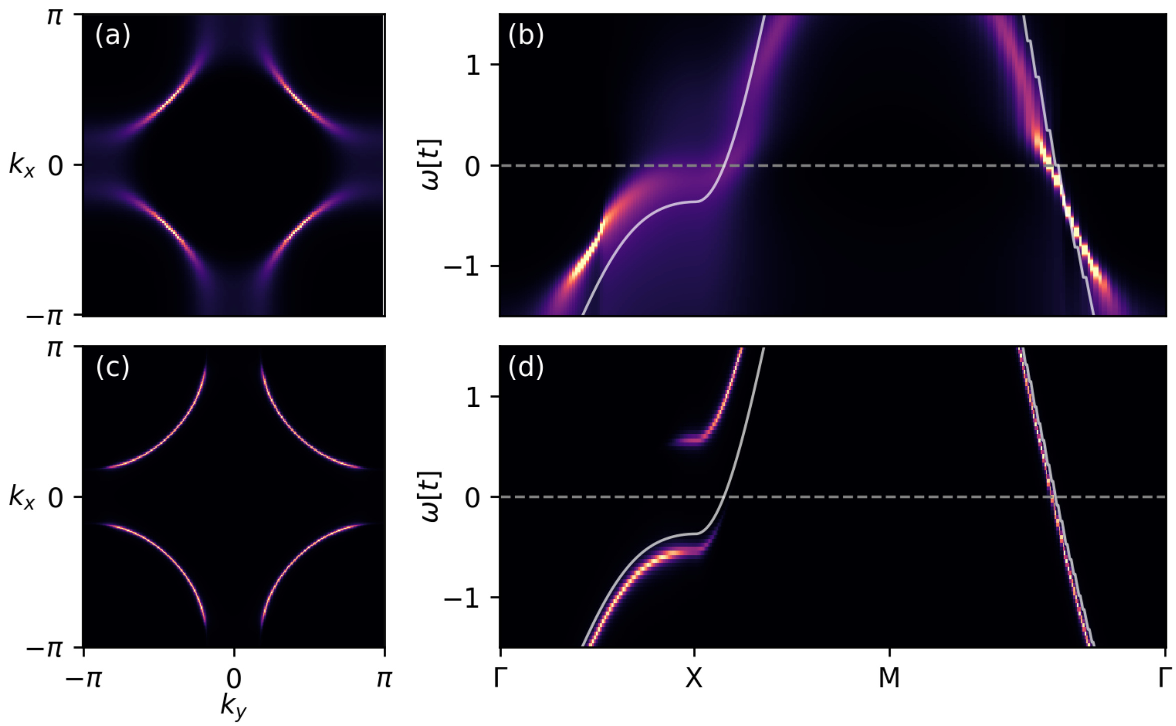


FIG. S12. (a,c) Comparison of the high-temperature spectral function at zero frequency $A(\omega = 0, k)$; and (c,d) of the spectral function $A(\omega, k)$ along the high-symmetry path, $\Gamma = (0, 0)$, X= $(\pi, 0)$ and M= $(\pi, \pi)$, in the Brillouin zone for $n = 0.85$ between DΓA at $U = 8t$ (top row) and the model Hamiltonian at $\mathcal{V} = 1.1t$ (bottom row). Temperature is $\beta = 7.5t$ which is somewhat above the pseudogap temperature, where there is a stronger broadening but not a gap opening at the antinode in DΓA.

## IX. HIGH TEMPERATURES

As mentioned right before the Conclusion in the main text, if one keeps the interaction strength $\mathcal{V}$ fixed to an appropriate value (e.g., in our case, $\mathcal{V} = 1.1t$) that describes the AF correlations in the pseudogap regime of the Hubbard model, it cannot be expected that the overall good agreement with the DΓA spectral functions found at low-to-intermediate temperatures also holds in the high-$T$ regime. We exemplify this effect by showing the comparison of the DΓA ($U = 8t$) and model ($\mathcal{V} = 1.1t$) spectral functions at $\beta = 7.5/t$ in Fig. S12: Here we observe that while the momentum-selective effects of DΓA, driven by nesting and AF spin fluctuations become gradually suppressed at higher temperatures, the built-in momentum selectivity of the model Hamiltonian represents an intrinsic property of its physics. In particular, a sharp gap remains always present in the spectral function of the model Hamiltonian, while the two corresponding “bands” become gradually more filled with higher temperatures. In order to describe the temperature dependence of the spectra of the Hubbard model more accurately, one could introduce a temperature-dependent interaction $\mathcal{V}(T)$ which decreases with temperature, becoming negligible above the pseudogap regime.

## X. INTERACTION DEPENDENCE OF THE MODEL HAMILTONIAN

In Fig. S13 we display the interaction $\mathcal{V}$ dependence of the spectral function, Fermi and Luttinger surfaces for the model Hamiltonian in the nodal Fermi arc regime (cf. Fig. 1 in the main text) at inverse temperature $\beta = 12.5/t$ and $n = 0.85$. For the $A(\omega, k)$ cuts along the Fermi-Luttinger surface (bottom row panels), at the Luttinger surface (gaped spectrum), the position in $\omega$ of the lower peaks is almost unaffected by an increase in $\mathcal{V}$, though its spectral weight gets reduced. The position of the higher second peak, on the contrary, gets shifted to higher values of $\omega$ with increasing spectral weight. This shift of spectral weight at the Luttinger surface results in a slightly reduced length of the Fermi arc for increasing values of $\mathcal{V}$ (top row panels).

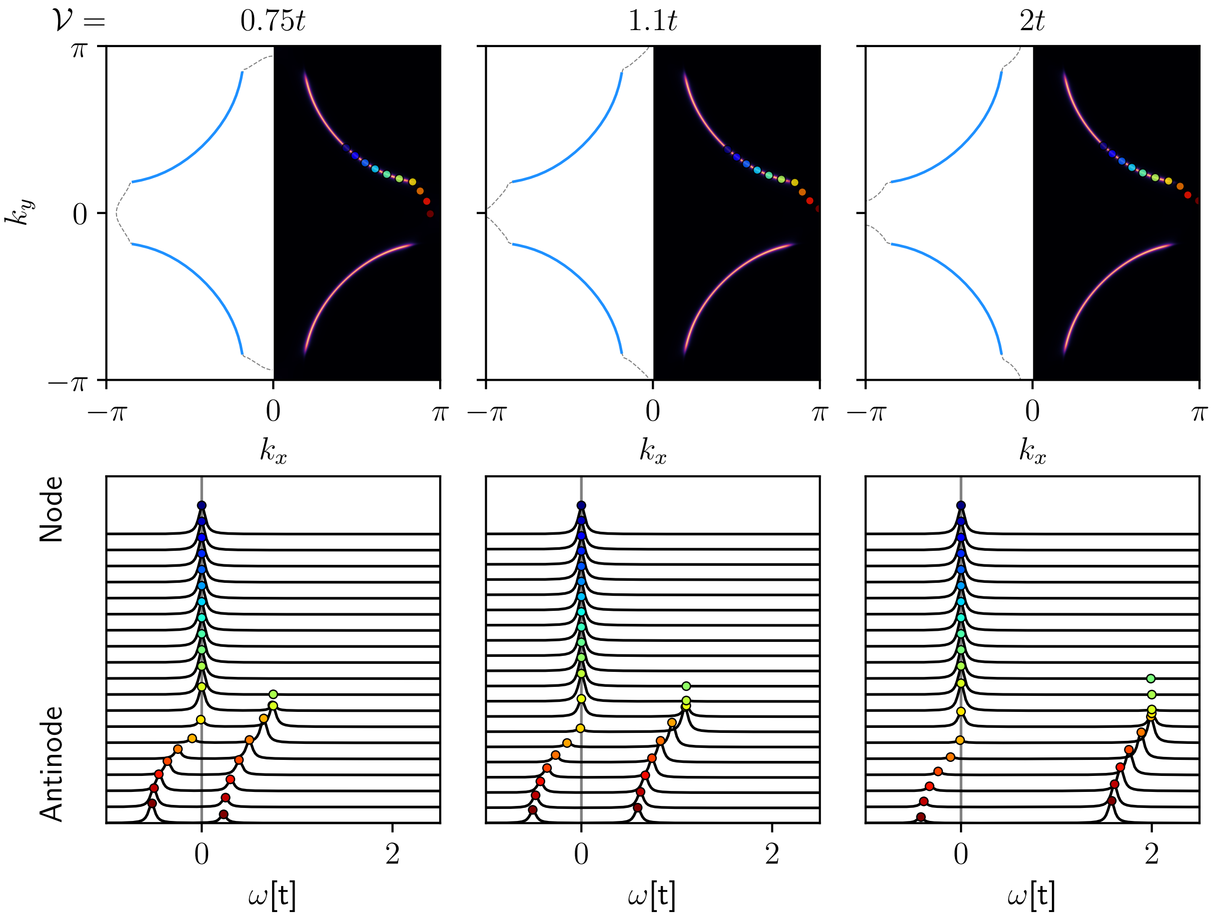


FIG. S13. Dependence on the model interaction value, $\mathcal{V}$, of the results Fig. 2 and Fig. 4 of the main text. Columns from left to right: $\mathcal{V} = 0.75t$, $1.1t$, $2t$; $\beta = 12.5/t$ and $n = 0.85$. Top row: Fermi (blue line), and Luttinger surfaces (gray dashed line) in the left half of the Brillouin zone; spectral function at zero frequency $A(\omega = 0, k)$ in the right half of the Brillouin zone. Colored dots mark the locations for the cuts in the spectral function below. Bottom row: $A(\omega, k)$ cuts along the Fermi-Luttinger surface for the momenta indicated by the color above.


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